Rajat Verma

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Professional Summary

Lead Data Scientist with 8+ years of experience building customer-level propensity models, revenue and CLTV-impact frameworks, and scalable ML pipelines on Google Cloud Platform. At The Home Depot, delivered production propensity and classification models, per-customer profitability impact analysis, and high-value customer acquisition scoring. Architected a YAML-driven GCP ML platform adopted by 10+ data scientists, compressing deep learning experimentation from months to under one week. Demonstrated business impact: 3.7% incremental lift in customer acquisition, 15.3% CTR improvement, greater than 99% validated causal lift measurement, and $150K+ in quantified savings. Experienced in causal inference, XGBoost and deep learning classification, customer behavioral and lifetime value modeling, and MLOps on GCP.

Programming Skills

- **ML/DL:** PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, FAISS, HuggingFace Transformers
- **NLP/GenAI:** LLMs (Gemini, Gemma, Pegasus), Fine-tuning, RAG, Prompt Engineering, LangChain, Agentic AI, Multi-Agent Systems
- **Cloud & MLOps:** GCP (Vertex AI Pipelines, BigQuery, Cloud Composer, GCS), Apache Airflow, Docker, MLflow, CI/CD
- **Data & Languages:** Python, SQL, BigQuery, Pandas, PySpark, Spark, R, Flask, FastAPI, Streamlit, Git

Work Experience

The Home Depot (via Decision Culture) — Remote, India

Lead Data Scientist — *February 2022 – Present*

Tech Stack: Python, GCP (Vertex AI Pipelines, BigQuery, Cloud Composer), Apache Airflow, PyTorch, FAISS, MLflow

- Recognized with the **Orange Hand Award (x2)** for outstanding individual contribution.
- Architected a YAML-driven Python framework that orchestrates the full ML lifecycle (feature generation, model training, and batch scoring) across GCP services including BigQuery, Vertex AI Pipelines, and Cloud Composer. Reduced deep learning experimentation from months to under one week. Adopted by 10+ data scientists, powering 10+ models with 3 in production testing. Currently integrating MLflow for experiment tracking and model registry.
- Spearheaded Home Depot's transition from siloed per-class XGBoost propensity and classification models to a standardized deep learning stack (LSTM and Transformer), with every migrated model outperforming its predecessor. Established a scalable, repeatable paradigm for customer behavior prediction and propensity scoring across the organization.
- Built a reusable causal inference framework on GCP to quantify per-customer revenue and CLTV impact where A/B testing is infeasible, including profitability impact analysis of email opt-out events to inform targeted retention and revenue preservation strategies. Validated at greater than 99% alignment against a live A/B test, establishing it as a trusted methodology for customer-level financial impact measurement across cross-functional teams.
- Developed a FAISS-based customer propensity and lookalike scoring model on Vertex AI to identify and rank high-value prospect customers by acquisition likelihood, delivering 3.7% incremental lift over the existing baseline. Batch inference orchestrated via Cloud Composer for scalable production scoring.
- Built an automated BigQuery SQL pipeline using reusable templates for feature engineering, generating TensorFlow-compatible training datasets and eliminating manual query writing across the team.
- Conducted an internal workshop on Agentic AI Workflow Fundamentals for data science peers, building and open-sourcing a multi-agent Natural Language to SQL system as educational material. Demonstrated secure multi-LLM orchestration (Gemini, Gemma via OpenRouter), guardrail-based input validation, schema introspection, and Streamlit deployment, upskilling the team on production-grade AI agent architecture.
- Mentor and upskill data scientists on deep learning training diagnostics, production code standards, and architectural best practices. Lead code reviews and oversee integration quality across the team repository.

Times Internet Delhi, India
Data Scientist *December 2020 – February 2022*

- Deployed a personalized news recommendation pipeline serving 1.5M Economic Times subscribers, lifting click-through rate from 4.8% to 15.3%. Built end-to-end orchestration via Apache Airflow for scheduling, monitoring, and logging.
- Fine-tuned Google's Pegasus for financial news summarization on ET Markets data, an early production LLM application, and shipped it as a Flask API currently serving live traffic.
- Designed a two-stage keyword recommendation system for editorial articles: BM25 retrieval over Wikipedia-indexed Solr followed by BERT embedding reranking, improving content tagging and discoverability.
- Hack and Hustle 3.0 Finalist (2022): Led a 3-person team to build a real-time Twitter social listening API delivering per-stock sentiment, subjectivity, and engagement analytics.

ZS Pune, India
Data Science Associate *July 2018 – December 2020*

- Automated rare disease identification from Electronic Health Records using a GAN-based PU-Classifier, compressing a multi-week manual process to 10 work hours.
- Built a resource-level propensity model predicting non-compliance likelihood for a Fortune 100 Hi-Tech client. Model-driven proactive interventions reduced non-compliance by 23% and delivered $150K in savings within 6 months.
- Modeled asymptomatic NASH progression from EHR data (86% AUROC), identifying 4 novel non-invasive markers for fast-progressing patients. Adopted for clinical trial enrollment prioritization and focused treatment.
- Mined unstructured clinical trial text to surface competitive landscape insights and research trend analysis for therapeutic area strategy.

ZS Pune, India
Data Science Intern *January 2017 – June 2017*

- Built a multi-class sentiment analysis pipeline on noisy social listening data, benchmarking bidirectional LSTM, TF-IDF Random Forest, and Word2Vec MaxEnt, achieving 72% accuracy and 0.45 Cohen's Kappa.

Education

Indian Institute of Technology Kharagpur West Bengal, India
Dual Degree (B.Tech and M.Tech) in Industrial Engineering *April 2013 – March 2018*

Certifications

- **Generative AI Leader Professional Certificate** – Google Cloud (Coursera), October 2025 (Verify Credential)

Projects

- **Multi-Agent SQL Generator:** Developed an end-to-end NLP system orchestrating multiple LLMs (Gemini/Gemma) for query classification, schema introspection, and SQL generation. Integrated strict validation guardrails to prevent SQL injection and deployed via a Streamlit interface.
- **Screen Scape** (rajatverma212/ScreenScape)**:** Client-side web tool using React, Vite, and Tailwind CSS with daisyUI, enabling users to visualize and compare computer monitor sizes interactively in real-world scale.
- **Sequential Recommender in Production – RecBole** (rajatverma212/recbole_to_production)**:** Comprehensive guide and implementation for deploying RecBole's Sequential Models (SHAN, SINE) as production-ready FastAPI endpoints, bypassing RecBole's experimentation-focused interfaces to serve recommendations using only item history without user IDs.
- **Motion Following Camera:** Built using Arduino Uno and IR sensors for autonomous object tracking.